

Proof. We show that there exists $0 < t_0 < b$ such that the hypothesis of Theorem 2.14 hold for the sequence of flows $\tilde{g}_i(t) = g_i(t_0 + t)$ for $t \in (-t_0, b - t_0)$.

Since $\|\text{Rm}(g_i(t))\| \leq C$ for all $t \in [0, b]$, then by Proposition 2.16 there exists a constant $K := n\sqrt{C} \geq 0$ such that for all $i \in \mathbb{N}$ we have for $t \in [0, b]$ we have

$$(1) \quad e^{-2Kt} g_i(0) \leq g_i(t) \leq e^{2Kt} g_i(0).$$

Fix $p \in M$, and consider two orthonormal vectors $u, v \in T_p M$. Let $\{e_1, e_2, \dots, e_n\}$ be an orthonormal basis of $T_p M$ with $e_1 = u, e_2 = v$. Then we have that

$$\begin{aligned} |\text{Sec}(g_i(t))(u \wedge v)|^2 &= (\text{Sec}(g_i(t))(u \wedge v))^2 \\ &= (\text{Rm}(g_i(t))(u, v, u, v))^2 \\ &= \text{Rm}(g_i(t))_{1212}^2(p) \\ &\leq \sum_{i,j,k,\ell=1}^n \text{Rm}(g_i(t))_{ijkl}^2(p) \\ &= \|\text{Rm}(g_i(t))\|^2(p) \leq C^2. \end{aligned}$$

Thus there exists $\Delta > 0$ such that $|\text{Sec}(g_i(t))| \leq \Delta$ for all $i \in \mathbb{N}$ and $t \in [0, b]$.

Fix $t_0 \in (0, b)$, and consider a C^1 -continuous curve $\gamma_i: [0, 1] \rightarrow M_i$. We have the following relation between the length of γ_i with respect to $g_i(0)$, denoted by $\ell(g_i(0))(\gamma_i)$, and the length of γ_i with respect to $g_i(t_0)$, denoted by $\ell(g_i(t_0))(\gamma_i)$:

$$\begin{aligned} \ell(g_i(0))(\gamma_i) &= \int_0^1 \sqrt{g_i(0)(\gamma'(s), \gamma'(s))} ds \\ &\geq \int_0^1 \sqrt{e^{-2Kt_0} g_i(t_0)(\gamma'(s), \gamma'(s))} ds \\ &\geq e^{-Kt_0} \int_0^1 \sqrt{g_i(t_0)(\gamma'(s), \gamma'(s))} ds \\ &= e^{-Kt_0} \ell(g_i(t_0))(\gamma_i). \end{aligned}$$

From this, it follows that for all $i \in \mathbb{N}$ and arbitrary $x_i, y_i \in M_i$ we have

$$d_{g_i(0)}(x_i, y_i) \geq e^{-Kt_0} d_{g_i(t_0)}(x_i, y_i).$$

For p_i fixed and $r > 0$, let $q_i \in B_{e^{-Kt_0}r}^{g_i(0)}(p_i)$. Then since $e^{-Kt_0}r > d_{g_i(0)}(p_i, q_i)$, it follows that

$$r > e^{Kt_0} d_{g_i(0)}(p_i, q_i) \geq d_{g_i(t_0)}(p_i, q_i),$$

That is the open ball $B_{e^{-Kt_0}r}^{g_i(0)}(p_i)$ of radius $e^{-Kt_0}r$ centered at p_i with respect to $g_i(0)$ is contained in the open ball of radius r centered at p_i with respect to $g_i(t_0)$, i.e.

$$B_{e^{-Kt_0}r}^{g_i(0)}(p_i) \subset B_r^{g_i(t_0)}(p_i).$$

Observe that since $t_0 \in (0, b)$ and $K > 0$, then we have $1 \geq e^{-Kt_0} \geq e^{-Kb}$, and $1 \geq e^{-Knt_0} \geq e^{-Knb}$. Choose $2r_0 = \min\{\nu_0, 2\}$. Then we have that $2e^{-Kt_0}r_0 \leq 2r_0 \leq \nu_0 \leq \text{Injrad}(g_i(0))$ for all $i \in \mathbb{N}$. Then by [12, Proposition 14], there exists a constant $A(n)$, which depends only on the dimension n , such that:

$$\text{vol}(g_i(0))\left(B_{e^{-Kt_0}r_0}^{g_i(0)}(x_i)\right) \geq \frac{A(n)}{n^n} r_0^n e^{-Knt_0}.$$

for each $i \in \mathbb{N}$ and all $x_i \in M_i$.